

ECN 164 International Finance — Practice Mock Exam B

Questions · Unofficial self-study practice based on Assignment 1, Part I (Chapter 2: quotes, cross rates, bid–ask spreads, triangular arbitrage)

Instructions. 15 short-answer problems, 100 points, suggested time 90 minutes. Show your work. Round exchange rates to 4 decimal places (more when a rate is very small) and money amounts to 2 decimal places. A calculator is allowed.

Name: _____ Score: _____ / 100

1. (5 points) Flipping a quote. On October 2, 2026 the New Zealand dollar was worth about USD0.5850/NZD and the Swiss franc about USD1.2500/CHF.

- How many New Zealand dollars does one U.S. dollar buy (NZD/USD)?
- How many Swiss francs does one U.S. dollar buy (CHF/USD)?
- Which currency is the “base” currency in each of the original quotes?

2. (5 points) Paying in a foreign currency. You are traveling in Mexico and your hotel costs MXN2,400 per night. The exchange rate is MXN18.50/USD.

- What is the cost of one night in dollars?
- A week later the rate is MXN18.20/USD. Did the peso appreciate or depreciate against the dollar? What does the same room cost in dollars now?

3. (5 points) Lion City Electronics, Inc. needs to buy 3,000,000 Singapore dollars (SGD) to pay its Singaporean supplier. Its banker quotes bid–ask rates of SGD1.2840–1.2850/USD. What will be the dollar cost of the SGD3,000,000?

4. (7 points) Rates of change. The dollar was CHF0.8050/USD on September 25, 2026 and CHF0.7930/USD one week later.

- By what percentage did the U.S. dollar appreciate or depreciate against the Swiss franc?
- By what percentage did the Swiss franc appreciate or depreciate against the dollar? (Hint: first express the rates as USD/CHF.)
- Why are your answers to (a) and (b) not exactly the same size?

5. (8 points) Cross rates. Use the following quotes against the U.S. dollar: USD1.33/GBP, USD0.585/NZD, INR88.20/USD, MXN18.40/USD.

- Calculate the cross rate between the New Zealand dollar and the pound, in NZD/GBP and in GBP/NZD.
- Calculate the rupee–pound cross rate (INR/GBP).
- Calculate the peso–New Zealand dollar cross rate (MXN/NZD).
- With 10 currencies, how many exchange rates would dealers need to quote if every pair traded directly? How many if every currency traded only against the dollar? What does this tell you about the role of the dollar?

6. (7 points) Bid–ask spreads in the interbank market. A dealer quotes the Australian dollar at USD0.6548–0.6551/AUD.

- Which rate is the bid and which is the ask? Which rate do you get if you buy Australian dollars from the dealer? If you sell Australian dollars to the dealer?
- What is the spread in pips? What is the spread as a percentage of the ask price?
- A company buys AUD5,000,000 from the dealer and, after changing its mind, sells them back to the dealer a minute later at the same quotes. How many dollars does the company lose?

7. (6 points) Bid–ask spreads at the airport. At the airport, a currency kiosk buys Canadian dollars at USD0.68/CAD and sells Canadian dollars at USD0.77/CAD.

- What is the percentage spread? Compare it with your answer to Problem 6(b).
- You exchange \$800 into Canadian dollars at the kiosk, then change your mind and exchange all of them back into U.S. dollars. How many U.S. dollars do you end up with?
- Give two reasons why spreads for cash (“physical exchange”) are so much larger than interbank spreads.

8. (5 points) Flipping a bid–ask quote. A bank quotes the dollar at MXN18.38–18.42/USD. What are the bank's bid and ask prices for the Mexican peso, quoted in USD/MXN? (Hint: remember that the bank always buys low and sells high.)

9. (6 points) Cross rates with bid–ask spreads. A bank quotes USD0.6548–0.6551/AUD and MXN18.38–18.42/USD. A customer wants to trade Australian dollars for pesos, and the bank will go through the dollar.

- What is the bank's bid rate for Australian dollars in terms of pesos (MXN/AUD)?
- What is the bank's ask rate for Australian dollars in terms of pesos (MXN/AUD)?

10. (6 points) A Mexican avocado exporter receives AUD2,000,000 from an Australian customer and converts it into pesos with the bank, using the quotes in Problem 9.

- Which MXN/AUD rate applies, and how many pesos does the exporter receive?
- How many fewer pesos does the exporter receive than it would at the mid-rates of both quotes?

11. (7 points) Cross rates over time. On September 1, 2026 the quotes were USD1.3400/GBP and INR87.50/USD. On October 1, 2026 they were USD1.3150/GBP and INR88.40/USD.

- Calculate the INR/GBP cross rate on both dates.
- By what percentage did the pound appreciate or depreciate against the rupee?
- By what percentage did the rupee appreciate or depreciate against the pound?

12. (7 points) As a foreign exchange trader, you see the following quotes for New Zealand dollars (NZD), U.S. dollars (USD), and Indian rupees (INR):

USD0.5850/NZD INR50.10/NZD INR88.20/USD

Is there an arbitrage opportunity, and if so, how would you exploit it? Compute your profit per \$1,000,000.

13. (10 points) Triangular arbitrage. You observe the following quotes (ignore bid–ask spreads): USD1.3300/GBP in New York, USD0.5850/NZD in Wellington, and NZD2.2200/GBP in London.

- What NZD/GBP cross rate is implied by the two dollar quotes?
- Is there an arbitrage opportunity? In which market is the pound “too cheap”?
- You start with \$1,000,000. Describe the three trades you would make and calculate your profit in dollars.
- What will happen to the three exchange rates as many traders do the same thing?

14. (10 points) Triangular arbitrage with bid–ask spreads. Now the quotes are: USD1.3298–1.3302/GBP, USD0.5848–0.5852/NZD, and NZD2.2640–2.2660/GBP. Starting with \$1,000,000, check both possible routes (dollars → New Zealand dollars → pounds → dollars, and dollars → pounds → New Zealand dollars → dollars). Is there an arbitrage profit once you pay the bid–ask spreads?

15. (6 points) The foreign exchange market. Answer briefly.

- What is the interbank market, and who are its main participants?
- A customer wants to convert New Zealand dollars into Indian rupees. Why does the bank usually route the trade through the U.S. dollar even though the customer then pays two spreads?
- What is “liquidity,” and how is it related to the size of bid–ask spreads?